

FERREL RASHAD AKEYLA

Crafting high-performance web systems, interactive physics-driven experiences, and scalable data solutions. Selected works, technical architecture, and engineering showcase.

Hi, I'm Ferrel — a software engineer and creative technologist focused on crafting fluid web interfaces, real-time data visualisations, and interactive digital experiences.

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"CODE WITH PRECISION, DESIGN FOR PEOPLE."

As an undergraduate Computer Science student at Universitas Indonesia, my work bridges rigorous systems engineering with high-fidelity interaction design. Whether building high-frequency WebSockets telemetry engines, training quantitative machine learning models, or crafting 60 FPS Canvas visualisations, I strive for clarity, performance, and craftsmanship.

FULL-STACK SYSTEMS

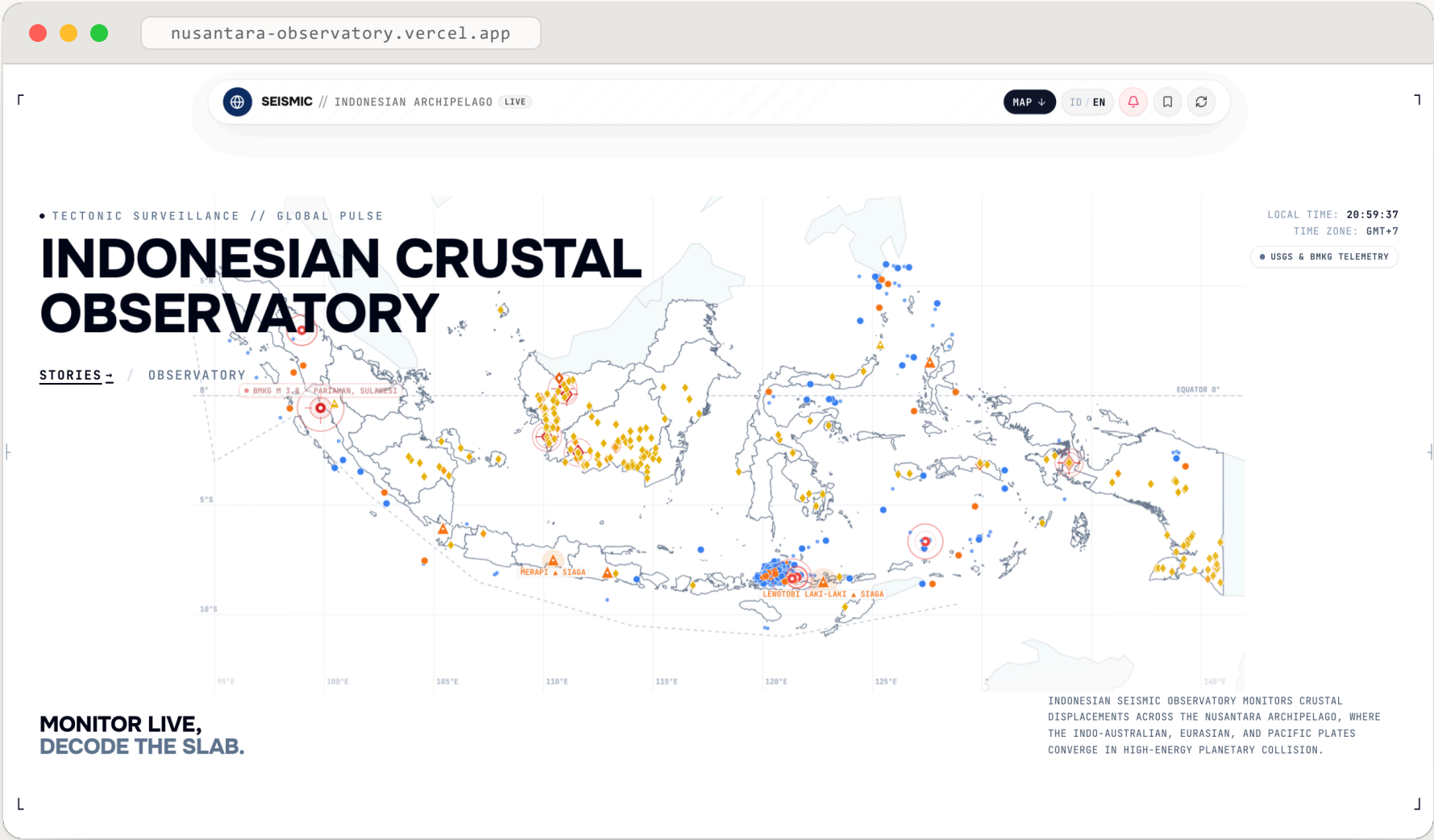
QUANTITATIVE ML

CREATIVE 3D & AUDIO



FERREL RASHAD AKEYLA

JAKARTA / DEPOK



NUSANTARA OBSERVATORY

ROLE: FRONTEND & GEOSPATIAL ENGINEER

SYSTEM: REACT 19 · TYPESCRIPT · CANVAS 2D · WEBSOCKETS · TAILWIND

DOMAIN: NUSANTARA-OBSERVATORY.VERCEL.APP

Hazard information is often scattered across different sources, making it difficult to see what is happening across Indonesia in one place. I turned fragmented hazard feeds into a single unified visual experience that makes complex geospatial events intuitive to explore.

I engineered an interactive geospatial observatory integrating multiple live data sources into a real-time 2D map with temporal event replay, fault line vector overlays, and proximity radius calculation. This project required combining high-performance data pipelines, WebGL/Canvas rendering, and responsive interaction into one cohesive system.

PARTICLE REPLAY ENGINE BMKG / USGS MULTI-INGESTION

60 FPS

CANVAS 2D ENGINE

< 50ms

PROXIMITY ANALYSIS

1,000+

SEISMIC EVENTS

Live Feed

BMKG & USGS INGESTION

tactiq.analytics

VE SCORES

FT LIV 3-1 CHE

21:00 AVL vs TOT

FT BAY 3-2 PSG

23:45 RMA vs BVB

HT BAR 0-0 RMA

EDS & FILTERS

All Matches24

Live Matches2 LIVE

Match Center→

PULAR LEAGUES

Pinned

Premier League8

Champions League6

La Liga5

Serie A4

Bundesliga4

Ligue 13

Eredivisie2

YesterdayTodayTomorrow

PREMIER LEAGUE · GAMEWEEK 8

Old Trafford

Man United7-0Man City

5th in PL2nd in PL

88'

Fernandes 14' (P), 78'

Rashford 28', 53'

Højlund 41'

Garnacho 65'

Mainoo 88'

xG 4.62 — xG 0.38

Match Center →

Premier League · England

Standings >

88'

Man United7

Man City0

FT

Liverpool3

Chelsea1

Premier League Table

#	CLUB	P	GD	P
1	Arsenal	8	+18	
2	Man City	8	+16	
3	Liverpool	8	+14	
4	Aston Villa	8	+8	
5	Chelsea	8	+7	

Top Scorers

1	Erling Haaland	14
2	Mohamed Salah	11
3	Bukayo Saka	9
4	Cole Palmer	8

60 FPS

DOUBLE-BUFFERED CANVAS

< 45ms

SOCKET.IO TRANSPORT

7 Axes

MULTIVARIATE RADAR

10K+ / match

TRACKING COORDINATE VECTORS

TACTIQ

ROLE: FRONT-END LEAD & UI ENGINEER

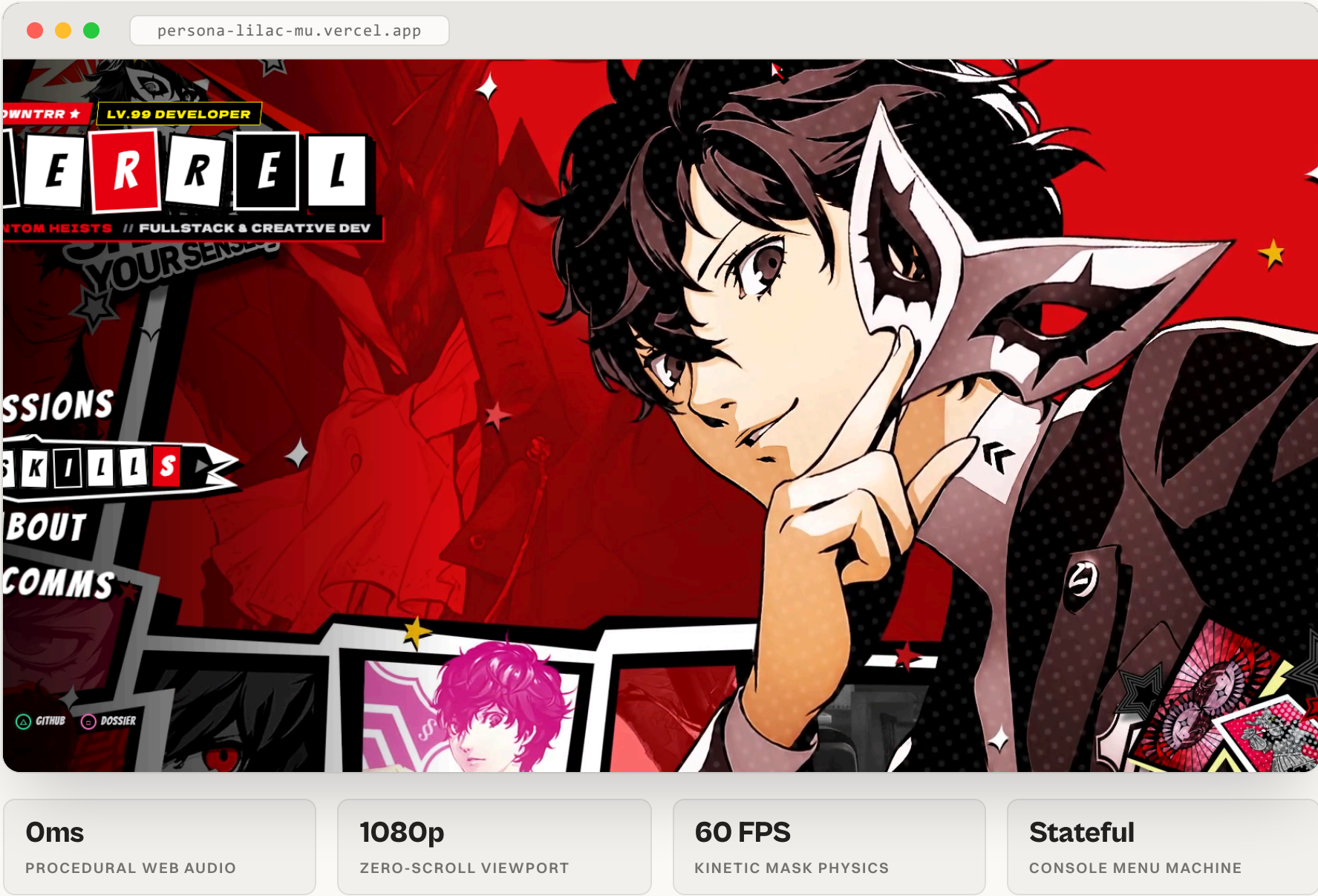
SYSTEM: NEXT.JS 14 · TYPESCRIPT · HTML5 CANVAS · SOCKET.IO · CHART.JS · TAILWIND

CONTEXT: TELEMETRY ENGINEERING & FOOTBALL ANALYTICS PLATFORM

Modern football coaching staff and analysts face an information overload: telemetry data streams, raw tracking coordinates, and delayed video recordings exist in fragmented silos. Analysts spent hours reconciling coordinate feeds without real-time spatial vectors.

As Front-End Lead, I architected the core visualization pipeline in Next.js 14. We isolated high-frequency coordinate broadcasts from React DOM reconciliations by utilizing a double-buffered HTML5 Canvas 2D engine driven by requestAnimationFrame. Synchronized linear interpolation prevents packet jitter, rendering live pitch heatmaps and passing corridors smoothly.

PITCH HEATMAPS SIMILARITY ENGINE



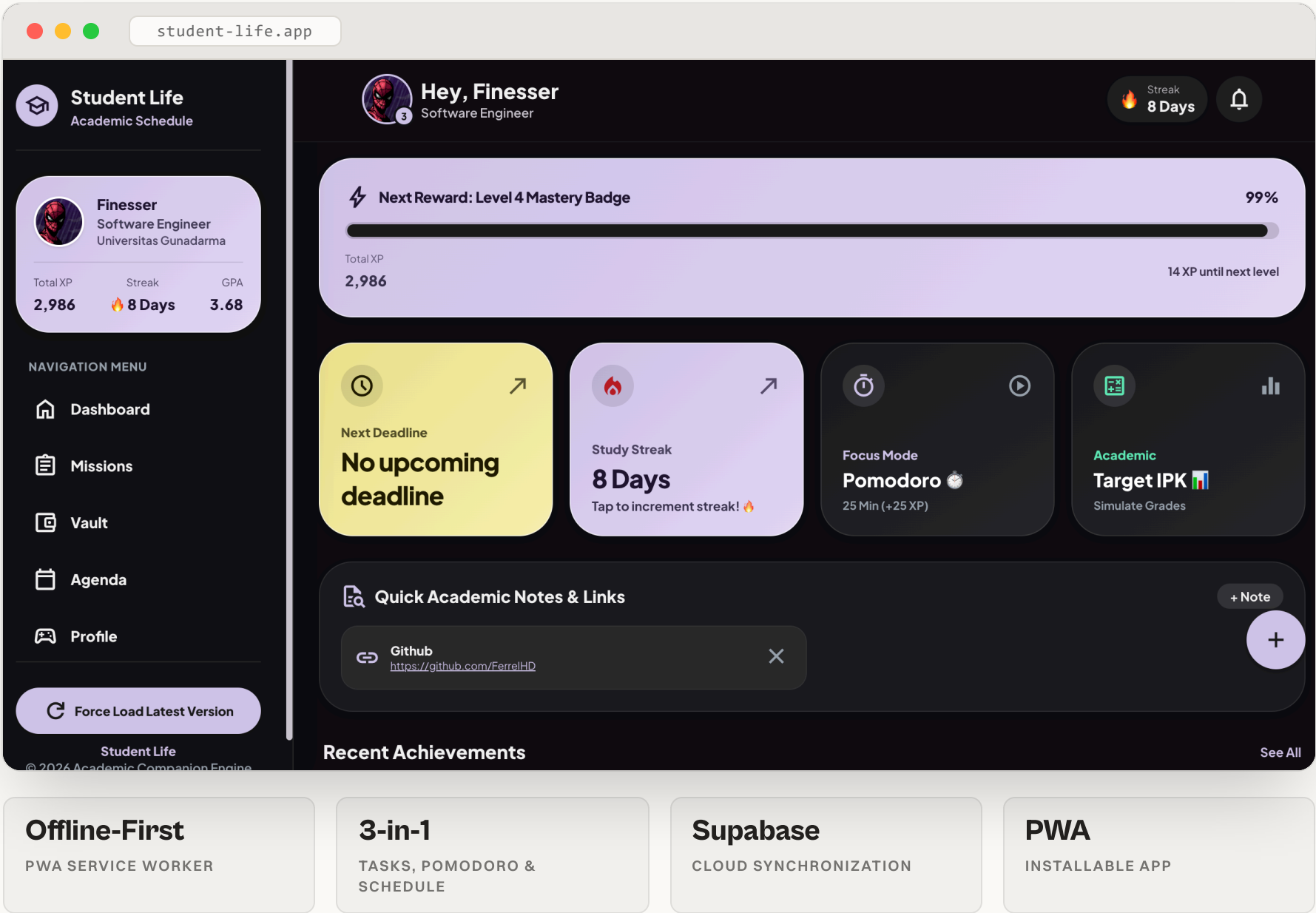
PERSONA 5 ROYAL

ROLE: CREATIVE DEVELOPER & UI ENGINEER
SYSTEM: REACT 18 · TYPESCRIPT · WEB AUDIO API · TAILWIND CSS
DOMAIN: PERSONA-LILAC-MU.VERCEL.APP

Most developer portfolios follow identical grid patterns and neutral aesthetics. I wanted to break the mold entirely by building an authentic, 60 FPS console experience inspired by Persona 5 Royal that pushes web interactivity to its absolute limit.

I engineered a zero-scroll 1080p viewport interface featuring procedural Web Audio synthesis for zero-latency acoustic feedback, ransom-note comic typography, character background video loops, and responsive game-menu state machines that recreate the iconic ATLUS experience right in the browser.

SYNTHESIZED SOUNDSCAPES ANGULAR POLYGON MASKS



STUDENT LIFE

ROLE: LEAD DEVELOPER & PRODUCT DESIGNER
SYSTEM: REACT 19 · TYPESCRIPT · SUPABASE · PWA
DOMAIN: STUDENT-LIFE.APP

Students juggle tasks, schedules, and focus sessions across multiple apps. I wanted to consolidate everything into one focused, offline-first productivity ecosystem tailored specifically for academic workflow.

I designed and developed an all-in-one student tool featuring smart task management, Pomodoro focus cycles, and schedule tracking with PWA offline support — so students can stay productive anywhere without network dependence.

POMODORO FOCUS CYCLES OFFLINE SERVICE WORKER

01

Frontend & Systems

High-speed web platforms with React 19, Next.js 14/15, Vue 3, TypeScript, and Tailwind v4. Type-safe state management, 60 FPS motion with GSAP, and modular architecture.

- REACT 19
- NEXT.JS 14
- VUE 3
- TYPESCRIPT
- GSAP MOTION
- TAILWIND V4
- CANVAS 2D

02

Creative Tech & 3D

Console-grade game UI architectures, procedural Web Audio synthesis, kinetic typography, and physics-driven 3D environments engineered for cinematic immersion.

- WEB AUDIO API
- UNITY 3D (C#)
- BLENDER 3D
- ROBLOX (LUAU)
- INTERACTIVE PHYSICS
- AUDIO SYNTHESIS

03

Full-Stack & ML

End-to-end architectures with Laravel 11, Node.js, Express, and PostgreSQL, coupled with quantitative ML pipelines for predictive intelligence and data analysis.

- PYTHON
- LARAVEL 11
- NODE.JS
- POSTGRESQL
- SUPABASE
- XGBOOST
- LIGHTGBM
- REST APIS

CONTACT & COLLABORATION

NEXT CHAPTER

Available for Software Engineering, Frontend/Full-Stack, and Creative Development roles.
Open for high-impact projects, collaborations, and engineering discussions.

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LINKEDIN

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LOCATION / BASE

Depok / Jakarta, Indonesia (GMT+7)